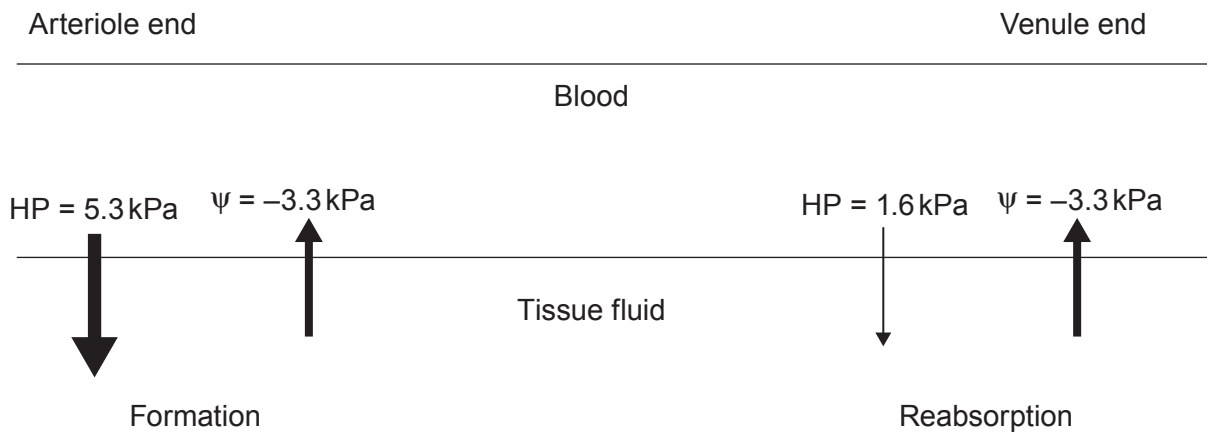


5. Tissue fluid is formed at the arteriole end of a capillary bed and water is reabsorbed at the venule end.

The formation of tissue fluid and reabsorption of water occurs as a result of two opposing forces: the hydrostatic pressure of blood and osmotic pressure. Proteins in the plasma maintain a low water potential in the blood.

The diagram illustrates the differences in hydrostatic pressure (HP) and water potential (ψ) between blood and tissue fluid in a capillary bed.



(a) Use the information in the diagram to calculate the pressure difference that results in:

(i) tissue fluid formation; [1]

..... kPa

(ii) water reabsorption. [1]

..... kPa

(iii) Name the vessel that removes excess tissue fluid. [1]

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(b) Explain why fluid will accumulate in the tissues of a person whose diet is poor in protein. [3]

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